PO-HAN LIN

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EDUCATION

National Taipei University of Technology — Taipei, Taiwan
Bachelor of Science in Information and Finance Management — Jun 2019 – Jun 2023
GPA: 3.42/4.0

EXPERIENCE

City Science Lab @ Taipei Tech (in collaboration with MIT Media Lab) — **Mar 2024 – Present**
Researcher (Full-time) — *Taipei, Taiwan*

- Architected a **Kubernetes-native** infrastructure for spatiotemporal data services, integrating automated ETL pipelines, data governance policies, and real-time service monitoring metrics.
- **Designed and implemented a unified Spatio-Temporal Database** adopting **H3 DGGS** to standardize heterogeneous spatial data into multi-resolution hexagonal grids, enabling consistent analysis across scales.
- Developed a high-performance web visualization platform using **Deck.gl**, **DuckDB** and **Apache Arrow**, capable of rendering over **300,000 geospatial entities** in real-time.
- Optimized client-server data transmission using **Apache Arrow** and **DuckDB**, achieving a **35% reduction in latency** and lowering large object serialization overhead.
- Engineered an end-to-end **Persona Simulation pipeline** that constructs behavioral profiles from video archives, utilizing **ASR** and **Speaker Diarization** for automated data extraction.
- **Established a video data extraction workflow** to fine-tune **Qwen 2.5/3** for text style transfer and leverage **BreezyVoice/Higgs-Audio** for realistic voice cloning.

City Science Lab @ Taipei Tech (in collaboration with MIT Media Lab) — **Mar 2023 – Oct 2023**
Undergraduate Research Opportunity Program (UROP) — *Taipei, Taiwan*

- Studied quantitative analysis on urban datasets to model complex causalities
- Collected data from open-source platforms and government agencies using **Apache Airflow** to orchestrate DAG workflows.
- Integrated **ChatGPT-3.5 Turbo** into Agent-Based Modeling to introduce **behavioral heterogeneity**, simulating a diverse array of citizen lifestyles and complex urban scenarios.
- Engineered and deployed a visualization layer using the **Deck.gl** framework, enhancing the platform's capability to render simulation results.

SELECTED PROJECTS

Insights Navigator | Python, React, Ollama, Milvus, PostgreSQL — **Apr 2024 – Jul 2024**

- Deploy localized models into a unified workflow for the Kaohsiung City Government, processing unstructured citizen hotline data to extract structured urban insights.
- Implemented **similarity search and graph visualizations** to uncover correlations between distinct topics (e.g., traffic congestion), integrating spatial mapping to help authorities locate problem origins.
- Integrated **ChatGPT-4o** to dynamically cluster petition issues and generate real-time summaries, enabling rapid identification of emerging urban problems and sub-issues.

Group Trading Strategy Portfolio Optimization | Python, Dask, YFinance — **Jul 2021 - Jul 2022**

- Outperformed the bear market benchmark -21% return) by mitigating losses to under 5%, utilizing a Grouping Genetic Algorithm to identify robust portfolio combinations against market volatility.
- Implementing parallel computing, significantly shortening the latency for large-scale strategy evolution.
- Developed an interactive visualization platform using React that enables users to configure parameters, visualize real-time backtesting results, and dynamically adjust portfolio strategies.

PUBLICATIONS

Democratizing Multi-Granularity Spatio-Temporal... | *ACM SIGSPATIAL (GeoGenAgent '25)* — **Nov 2025**

- Authors: Che-Cheng Wu, Syuan-Bo Huang, Yu-Lun Song, **Po-Han Lin**, et al.
- Designed the Multi-Agent System with a robust interaction pipeline between LLM agents and the H3 spatial database using LangGraph.

TECHNICAL SKILLS

Languages: Python, TypeScript, C++, SQL, Bash
Frameworks & Tools: Linux, Git, Kubernetes, Docker, Helm, LangGraph, OLlama, PostgreSQL, DuckDB